Exploring the Feasibility of Zero-Shot Super-Resolution in Preclinical Imaging

Reviewed and reproduced by Amine Maazizi and Nemanja Vujadinovic

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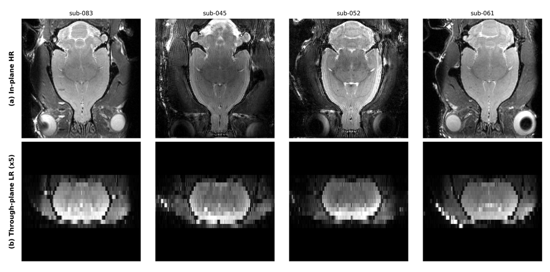
Motivation

Preclinical MRI is often **anisotropic**: scans have good in-plane resolution but poor through-plane resolution because slices are thick. This limits 3D tasks such as skull stripping, registration, and quantitative analysis.

The paper studies **zero-shot through-plane super-resolution** (SR) for rodent MRI. A major difficulty is that supervised SR needs paired LR/HR data and often generalizes poorly across protocols and species. The goal is therefore to recover HR structure **without paired target-domain supervision**.



(a) In-plane HR and (b) Through-plane LR view.

Positioning to the State of the Art

Three families of methods are relevant to this problem:

- **Interpolation**: simple, but cannot recover missing high-frequency detail [6].
- **Supervised SR**: effective when paired data exists, but sensitive to domain shift [1, 2].
- **Self-supervised MRI SR**: more flexible, but often slice-based and less consistent in 3D [4, 5].

The paper proposes a **zero-shot diffusion-based** alternative with **biplanar averaging** (BiDA).

Objective and Contributions

Objective: Reconstruct HR preclinical MRI volumes from anisotropic acquisitions without requiring paired target-domain supervision.

Main contributions:

- A biplanar, zero-shot diffusion reconstruction method for anisotropic preclinical MRI that combines axial and sagittal reconstructions to improve volumetric consistency.
- Adaptation of DDNM-style inference with DDPM-based HR image priors to perform SR without paired LR/HR training data.
- Evaluation at two SR scale factors on in- and out-of-domain rat data, and cross-species mouse data, using both reconstruction and downstream task metrics.

Our work:

- Critical review and analysis of the proposed method.
- Lightweight reproduction of the core BiDA approach.

Methodology

The method consists of four main components, which are summarized in the image below.

Degradation model. The reconstruction problem is formulated as recovering a HR image x from an LR observation y under the forward model

$$y = Ax.$$

Here, A represents the degradation operator consisting of blur and downsampling. Modeling this degradation allows the reconstruction to enforce consistency with the observed measurements.

DDPM training and priors. Two independent DDPMs are trained on HR slices for the **axial** and **sagittal** axes. These models learn a prior distribution of realistic HR slices, which later guides reconstruction without requiring paired supervision.

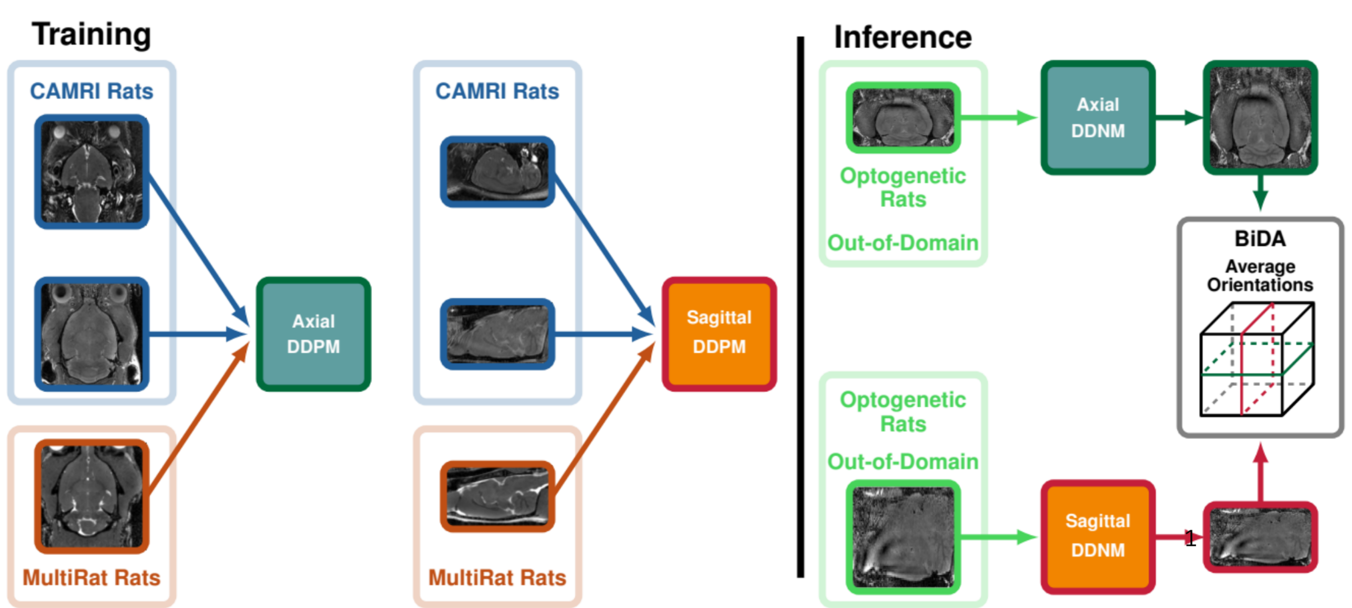
DDNM inference. During inference, HR images are reconstructed using DDNM sampling:

$$\hat{x}_{0|t} = A^{\dagger}\mathbf{y} + (I - A^{\dagger}A)x.$$

This decomposition preserves the LR observation while allowing the diffusion prior to estimate the missing high-frequency details. In addition, DDIM sampling is used to accelerate inference.

Averaging and final reconstruction. Reconstruction is performed independently along both axes, slice-by-slice, which are then stacked to obtain volumes $\hat{x}_{\text{axial}}$ and $\hat{x}_{\text{sagittal}}$. The final reconstruction is then obtained as the average of the two, in order to reduce the slice inconsistency:

$$\hat{x}_{\text{BiDA}} = \tfrac{1}{2}(\hat{x}_{\text{axial}} + \hat{x}_{\text{sagittal}}).$$



The overview of BiDA method.

Validation Protocol

Data. The method operates on several rodent MRI datasets: CAMRI rats and CAMRI mice, MultiRat, and Optogenetic rats. The datasets are split as follows:

- **Training**: Isotropic CAMRI rat and MultiRat scans.
- **Validation**: Anisotropic CAMRI rat scans.
- **OOD testing**: Optogenetic rat (held-out) and CAMRI mouse scans.

Procedure. First, LR inputs are created from HR volumes using slice-profile blurring and through-plane subsampling to simulate thick-slice acquisition. Through-plane SR is then applied and evaluated at $2.5\times$ and $5\times$ scale factors to recover the initial isotropic resolution.

Baselines. BiDA is compared against cubic interpolation and self-supervised SR method *ECLARE* [5].

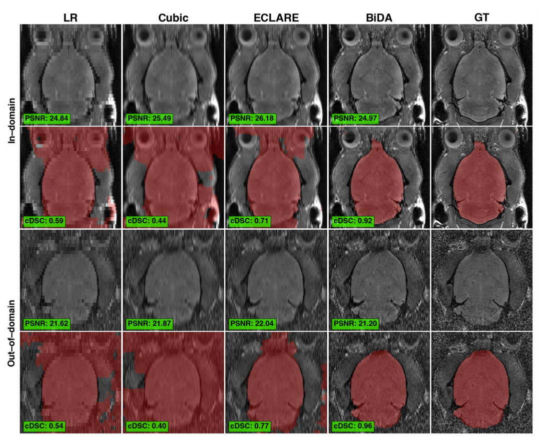
Metrics. Reconstruction quality is evaluated using PSNR and a downstream skull-stripping task is measured with cDSC.

Main Findings from the Paper

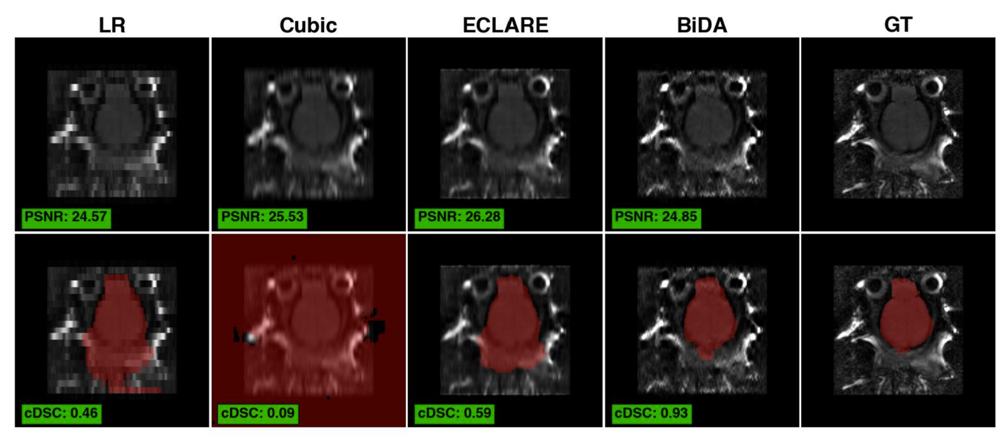
The results reveal several notable findings of the proposed method:

- **Perceptual quality**: BiDA produces sharper and more structurally plausible through-plane details in qualitative examples.
- **PSNR tradeoff**: *ECLARE* often achieves higher PSNR, which reflects the perception–distortion tradeoff where sharper perceptual reconstructions may not maximize PSNR metric.
- **Downstream task**: Despite lower PSNR, BiDA improves skull-stripping consistency with higher cDSC and fewer failure cases.
- **Generalization**: Both methods perform well on in-domain rat data, but BiDA shows weaker performance on cross-species mouse data.

Overall support: These results support the paper's main claim for perceptual quality and downstream utility, but only partially for strong generalization and for isolating the effect of biplanar averaging.



Rat results from the paper



Mouse results from the paper

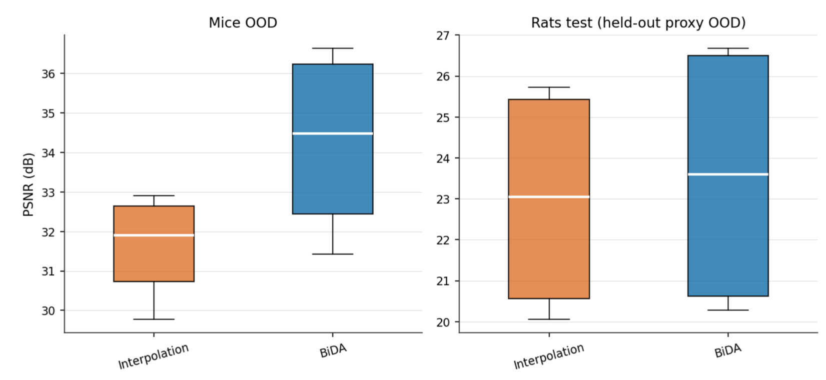
Qualitative, quantitative, and skull-stripping comparisons reported in the paper for rat and mouse settings.

Our Reproduction

In addition to reviewing the paper, we implement a lightweight reproduction of the BiDA method as a sanity check under limited compute. We kept the main ingredients of the method but simplified the setup by training only on **CAMRI rat** data, using **unconditional** training on HR slices, and evaluating a narrower benchmark with synthetic LR inputs and cubic interpolation as the main baseline.

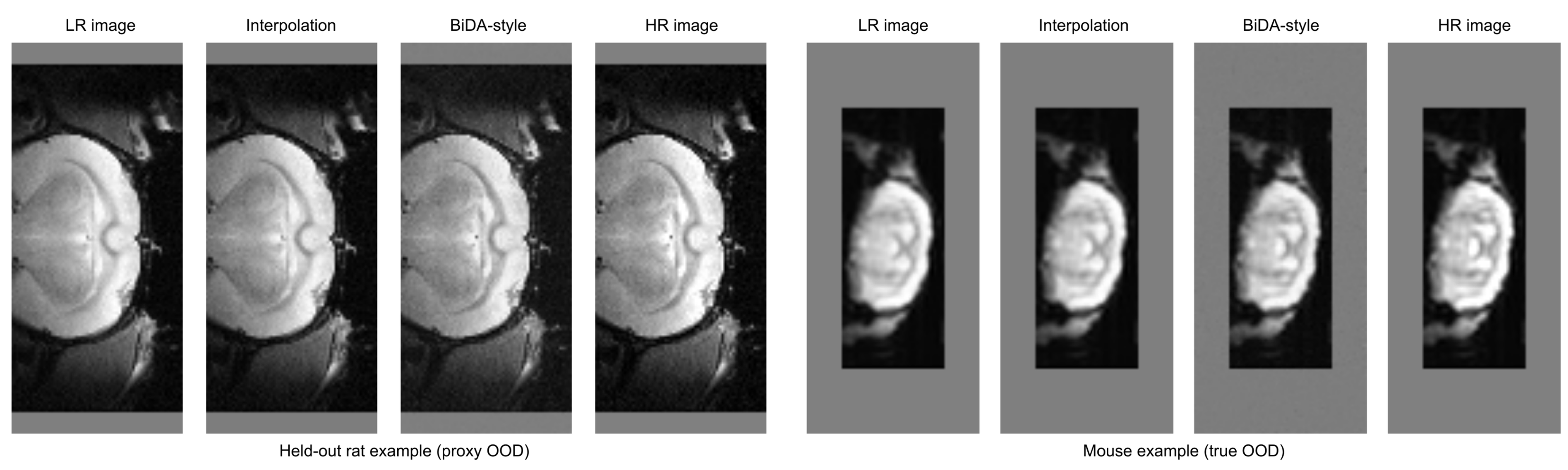
Quantitative results. Our simplified implementation improves **PSNR** on both datasets, but interpolation remains better on **SSIM** and **MAE**.

Dataset	Metric	BiDA	Interpolation
Mice OOD	PSNR	34.30 ± 1.96	31.66 ± 1.05
	SSIM	0.9316 ± 0.0107	0.9710 ± 0.0121
	MAE	0.0097 ± 0.0018	0.0071 ± 0.0019
Held-out Rats (proxy OOD)	PSNR	23.55 ± 3.02	22.96 ± 2.54
	SSIM	0.7792 ± 0.1153	0.8200 ± 0.1097
	MAE	0.0385 ± 0.0172	0.0367 ± 0.0154



PSNR comparison on the held-out rat and mouse sets.

Qualitative results. Our implementation yields sharper, more plausible reconstructions than interpolation on both rat and mouse sets, consistent with the perception–distortion tradeoff.



Qualitative comparison on one held-out rat and one mouse example.

Critical Assessment

Strengths

- Clinically relevant preclinical MRI problem with a simple, well-motivated biplanar design.
- Grounded in DDPM/DDNM/DDIM with evaluation beyond in-domain: held-out rats and cross-species mice.
- Includes downstream assessment (cDSC), not only reconstruction metrics.

Limitations

- No ablation of the biplanar design.
- cDSC relies on a silver-standard pipeline.
- LR inputs assume idealized no-slice-gap degradation.
- Missing comparison with *DiffusionMBIR* [3].
- Only one downstream task is evaluated.
- Mouse-domain failure remains unanalyzed.

Conclusion

BiDA frames anisotropic preclinical MRI SR as a **diffusion-based inverse problem**, avoiding paired supervision via zero-shot DDNM inference. Its claim on **perceptual quality and downstream utility** (cDSC) is well-supported; its claim that **biplanar averaging** drives the gains is not, due to missing ablations and the absent *DiffusionMBIR* comparison. Cross-species generalization also remains unconvincing. Our reproduction confirms the qualitative picture under limited compute.

Key insight. Cross-species failure is nontrivial, in inverse problems, even a small anatomical domain gap becomes a **null-space mismatch**: the diffusion prior must infer missing through-plane structure that lies outside its training support, unlike natural-image generation where visual plausibility suffices.

References

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